# SIMATS ENGINEERING
## ASSESSMENT TOOL 1

**COURSE NAME: CSA07 COURSE CODE: Computer Networks**

**COURSE OUTCOME COVERED**

***CO1:*** *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

*Assessment Tool Weightage: 40%*

Annexure A

**SHORT ANSWER TEST**

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***Code of Conduct:***

*I Y. Varun Kumar Reddy(192424230) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
*I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student:***

Annexure A

## Short Answer Test

1. A company needs to transfer a 150 MB design file from a client computer to a remote server over a network with a bandwidth of 50 Mbps. The Transport Layer divides the file into 1500-byte segments, while the Data Link Layer encapsulates each segment into frames by adding 40 bytes of header and trailer. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.

## Solution:

File size = 150 MB = 150,000,000 bytes Number of segments = 150,000,000 / 1500 = 100,000 segments Since each segment is encapsulated into one frame: Number of frames = 100,000 frames Size of each frame = 1500 + 40 = 1540 bytes Total Data Link Layer overhead = 100,000 × 40 bytes = 4,000,000 bytes = 4 MB (32,000,000 bits) Total data transmitted (with overhead) = 100,000 × 1540 = 154,000,000 bytes = 154 MB Total bits to transmit = 154,000,000 × 8 = 1,232,000,000 bits Transmission time = 1,232,000,000 bits / 50,000,000 bps = 24.64 seconds Explanation: The Transport Layer ensures reliable communication by segmenting the file into manageable units, numbering each segment for correct sequencing, using acknowledgements to confirm receipt, and retransmitting any segment that is lost or corrupted (e.g., as done by TCP). It also performs flow and congestion control so the sender does not overwhelm the receiver or the network. The Data Link Layer ensures reliability on each physical hop by framing the data, attaching error-detection codes (such as CRC) in the trailer, performing MAC-level addressing, and requesting retransmission of damaged frames between adjacent nodes. Together, these two layers guarantee that data arrives completely, in order, and without errors from source to destination

2. A Transport Layer protocol uses a sliding window size of 6 frames. Each frame carries 1024 bytes of user data. If the sender transmits 48 frames, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

## Solution:

Window size = 6 frames, Total frames to send = 48 Number of transmission windows = 48 / 6 = 8 windows If 1 frame is lost per window, retransmissions needed = 1 × 8 windows = 8 retransmissions Explanation: Flow control (implemented here through the sliding window mechanism) allows the sender to transmit multiple frames (a full window) before waiting for individual acknowledgements, instead of sending one frame and waiting for its acknowledgement each time. This keeps the communication channel busy, improves throughput and bandwidth utilization, and prevents a fast sender from overwhelming a slow receiver's buffer. When a frame is lost, only that frame (or window, depending on the protocol - Go-Back-N or Selective Repeat) needs retransmission, which keeps recovery efficient while still protecting the receiver from being flooded with unacknowledged data.

3. A 12,000-byte IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is 1500 bytes. Assuming each fragment requires a 20-byte IP header, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

**Solution:**

Maximum data (payload) per fragment = MTU − header = 1500 − 20 = 1480 bytes Total packet size = 12,000 bytes Number of fragments = ceil(12,000 / 1480) = ceil(8.11) = 9 fragments (8 fragments × 1480 bytes = 11,840 bytes; remaining 160 bytes go into the 9th fragment) Total header overhead = 9 fragments × 20 bytes = 180 bytes Explanation: The Network Layer ensures successful delivery of fragmented packets by assigning the same Identification number to all fragments of the original packet, using the Fragment Offset field to indicate each fragment's position within the original data, and using the More Fragments (MF) flag to mark all but the last fragment. Each fragment is routed independently across the network as a separate IP packet. At the destination, the Network Layer uses the identification number and offset values to reassemble the fragments in the correct order into the original packet before passing it up to the Transport Layer. If any fragment is lost, the entire original packet must be retransmitted

4. A Transport Layer protocol uses a sliding window size of 6 frames. Each frame carries 1024 bytes of user data. If the sender transmits 48 frames, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

**Solution:**

Window size = 6 frames, Total frames to send = 48 Number of transmission windows = 48 / 6 = 8 windows If 1 frame is lost per window, retransmissions needed = 1 × 8 windows = 8 retransmissions This question is identical to Question 2, so the working and results are the same. Explanation: Flow control (implemented here through the sliding window mechanism) allows the sender to transmit multiple frames (a full window) before waiting for individual acknowledgements, instead of sending one frame and waiting for its acknowledgement each time. This keeps the communication channel busy, improves throughput and bandwidth utilization, and prevents a fast sender from overwhelming a slow receiver's buffer. When a frame is lost, only that frame (or window, depending on the protocol - Go-Back-N or Selective Repeat) needs retransmission, which keeps recovery efficient while still protecting the receiver from being flooded with unacknowledged data.

5. A multimedia application transfers a **600 MB** video file. The Session Layer inserts a synchronization checkpoint after every **60 MB** of transmitted data. During transmission, a failure occurs after **390 MB** has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

**Solution:**

Checkpoints occur at: 60, 120, 180, 240, 300, 360, 420 MB, ... Failure occurs at 390 MB, which lies between the checkpoint at 360 MB and the next one at 420 MB Number of checkpoints created before failure = floor(390/60) = 6 checkpoints (last one at 360 MB) Data already confirmed at last checkpoint = 360 MB Data transmitted at time of failure = 390 MB Data to be retransmitted after recovery = 390 − 360 = 30 MB Explanation: Synchronization checkpoints in the Session Layer are vital because they allow a session to resume from the last confirmed checkpoint rather than restarting the entire transfer after a failure. This significantly reduces the amount of data that must be retransmitted (only 30 MB instead of the full 390 MB already sent), saving time, bandwidth, and processing resources. The Session Layer is responsible for establishing, maintaining, synchronizing, and gracefully recovering or terminating communication sessions between two systems, which is especially important for large file or multimedia transfers prone to interruption

6. Before transmission, the Presentation Layer compresses a **900 MB** file by **40%**. The compressed data is encrypted, adding **5%** overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

**Solution:**

Compressed size = 900 MB × (1 − 0.40) = 900 × 0.60 = 540 MB Encryption overhead = 5% of compressed size = 540 × 0.05 = 27 MB Encrypted (final) size = 540 + 27 = 567 MB Total reduction compared to original = 900 − 567 = 333 MB Total percentage reduction = (333 / 900) × 100 = 37% Explanation: The Presentation Layer improves transmission efficiency by compressing data before it is sent, which reduces the volume of data transmitted, lowers bandwidth consumption, and shortens transmission time, even after the encryption overhead is added back (a net 37% reduction here). It improves security by encrypting the data, ensuring confidentiality and protecting it from unauthorized access or eavesdropping during transit. The Presentation Layer also handles data translation and formatting (e.g., character encoding, data structure conversion) so that data sent by one system's application can be correctly interpreted by another, regardless of differences in internal data representation.

7. An FTP application transfers **3 GB** of data over a network with an effective throughput of **120 Mbps**. The Application Layer divides the data into requests of **3 MB** each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

**Solution:**

Total data = 3 GB = 3,000 MB = 3,000,000,000 bytes Number of requests = 3,000 MB / 3 MB = 1,000 requests Total bits to transmit = 3,000,000,000 bytes × 8 = 24,000,000,000 bits Transmission time = 24,000,000,000 bits / 120,000,000 bps = 200 seconds Explanation: The Application Layer provides reliable services to users by offering standardized protocols (such as FTP) through which user applications can request, send, and receive files without needing to manage low-level network operations. It breaks large transfers into smaller requests, which allows better error handling, progress tracking, and recovery from partial failures (only a failed request needs to be resent, not the whole file). It also relies on the underlying Transport Layer (TCP) for guaranteed, in-order, error-checked delivery, and it manages user-facing concerns such as authentication, session management, and presenting data in a usable form to the end user.

8. A packet experiences the following processing delays at different OSI layers: Application Layer = **4 ms**, Transport Layer = **3 ms**, Network Layer = **5 ms**, Data Link Layer = **2 ms**, and Physical Layer = **1 ms**. The propagation delay is **18 ms**, and the transmission delay is **12 ms**. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

**Solution:**

Total processing delay = 4 + 3 + 5 + 2 + 1 = 15 ms Total end-to-end delay = Processing delay + Transmission delay + Propagation delay Total end-to-end delay = 15 + 12 + 18 = 45 ms Explanation: Each OSI layer contributes to the total delay through the processing it performs: the Application Layer prepares and formats the user's data/request; the Transport Layer adds segmentation, sequencing and reliability information; the Network Layer performs routing and logical addressing decisions (often the largest processing delay, as seen here); the Data Link Layer performs framing, addressing, and error-detection code generation; and the Physical Layer converts the frame into raw bits for transmission. In addition to this cumulative processing delay, transmission delay (time to push all bits onto the link) and propagation delay (time for signals to travel the physical distance) add to the total end-to-end delay experienced by the packet.

**9.** A network transfers **80 MB** of useful data using frames of **1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers. Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

**Solution:**

Useful data per frame (payload) = 1600 − 80 = 1520 bytes Total useful data = 80 MB = 80,000,000 bytes Number of frames required = ceil(80,000,000 / 1520) = 52,632 frames Total overhead transmitted = 52,632 × 80 bytes ≈ 4,210,560 bytes (≈ 4.21 MB) Total data transmitted = 52,632 × 1600 bytes ≈ 84,211,200 bytes Transmission efficiency = (Useful data per frame / Total frame size) × 100 = (1520/1600) × 100 = 95% Explanation: Protocol overhead (headers/trailers added by various OSI layers) directly reduces the effective throughput of a network because part of every frame's capacity is consumed by control information rather than actual user data. This lowers transmission efficiency (here, 95%, meaning 5% of the bandwidth is consumed purely by overhead), increases the total number of frames and total bits that must be transmitted for the same amount of useful data, and consequently increases transmission time. Network designers must balance frame size and overhead carefully - very small frames waste more bandwidth on repeated overhead, while very large frames increase the impact of a single frame error.

**10.** A hospital transfers a patient's medical record of **240 MB** through an IoT-enabled healthcare network. The Presentation Layer compresses the data by **30%**, the Transport Layer divides the compressed data into **1400-byte segments**, and the Data Link Layer adds **60 bytes** of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

**Solution:**

Compressed size = 240 MB × (1 − 0.30) = 240 × 0.70 = 168 MB Compressed size in bytes = 168,000,000 bytes Number of segments = 168,000,000 / 1400 = 120,000 segments Data Link Layer overhead (60 bytes/frame, one frame per segment) = 120,000 × 60 = 7,200,000 bytes (7.2 MB) Final data transmitted across the Physical Layer = Compressed data + DLL overhead = 168,000,000 + 7,200,000 = 175,200,000 bytes (175.2 MB) Explanation: Multiple OSI layers work together to ensure secure and reliable delivery of the medical record: the Presentation Layer compresses (and would typically also encrypt) the sensitive patient data, reducing size and protecting confidentiality; the Transport Layer divides the compressed data into fixed-size segments, adds sequencing information, and ensures reliable, in-order, error-checked delivery, retransmitting lost segments if necessary; the Network Layer provides logical addressing and routing of each segment across the IoT healthcare network to the correct destination; the Data Link Layer frames each segment, adds error-detection information, and manages access to the physical medium between devices; and the Physical Layer transmits the final bits as electrical, optical, or radio signals. This layered cooperation ensures the medical data reaches its destination completely, accurately, securely, and efficiently despite the resource constraints typical of IoT devices.

## RUBRICS FOR CALCULATION

| Criteria | Weightage | Level 4 – Exemplary | Level 3 – Proficient | Level 2 – Developing | Level 1 – Beginning |
|---|---|---|---|---|---|
| Understanding of Concepts | 25% | Demonstrates a thorough, accurate understanding of concepts | Good understanding with minor gaps | Basic understanding with some misconceptions | Poor understanding; major misconceptions |
| Explanation | 25% | Detailed, well-explained answers with relevant examples | Clear explanation with adequate details | Limited explanation; lacks depth | Poor or unclear explanation |
| Application | 20% | Effectively applies concepts with strong analytical ability | Applies concepts with minor errors | Limited application with weak analysis | No application or incorrect analysis |
| Organization | 15% | Well-structured answers with logical flow and coherence | Organized with minor issues in flow | Basic structure, lacks clarity | Poor organization, incoherent answers |
| Accuracy | 10% | Completely accurate and covers all required points | Mostly accurate with minor omissions | Partially correct with missing points | Incorrect answers with major gaps |
| Clarity | 5% | Clear, neat, professional presentation | Understandable with minor issues | Limited clarity, readability issues | Poorly written, difficult to understand |